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Perspectives on research on laser driven proton-boron fusion and applications

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Perspectives on research on laser driven proton-boron fusion and applications

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ABSTRACT: Recent experiments with high-intensity lasers have shown record production of α -particles by irradiating boron-hydrogen targets. This opened the way to completely new studies on proton-boron fusion with multiple goals:

- i) studies related to nuclear fusion. The proton-boron fusion reaction produces 3 α -particles and releases a large energy. It is considered an interesting alternative to deuterium-tritium fusion because it produces no neutrons, therefore no activation and radioactive wastes.
- ii) generation of novel laser-driven α -particle sources. Laser-driven α -particle sources are promising for their potential high brightness while remaining compact. They could be used for multidisciplinary applications, including medical ones.

The COST Action CA21128 — PROBONO (PROton BORon Nuclear fusion: from energy production to medical applicatiOns) is the first international programme which aims at understanding the physics involved in laser-driven pB fusion, including the study of Equation of State of boron and boron compounds. Action's goals are to facilitate access to experimental infrastructures, maximize production of new knowledge, boost the career of young researchers by fostering opportunities for training, and finally interconnect researchers across countries building a well-organized community focused on pB research.

KEYWORDS: Interaction of radiation with matter; Nuclear instruments and methods for hot plasma diagnostics; Plasma diagnostics (remove charged particle spectroscopy); Plasma generation (laser-produced, RF, x ray-produced); Laser driven fusion; Proton-boron fusion; Laser driven particle sources, alpha particle, radioisotopes for medical applications



Contents

1	Introduction	1
2	State of art of the proton boron fusion experiment	2
3	Current challenges	5
3.1	Challenge on development of novel diagnostics	5
3.2	Challenge on targetry	7
3.3	Challenge on development on adapted numerical simulation tools	7
4	Summary	8

1 Introduction

During more than half of the last century, deuterium tritium fusion, leading to production of the alpha particle and highly energetic neutron (14,07 MeV), has been extensively studied both using the Magnetic Confinement Fusion, MCF, approach (especially on tokamak devices like JET [1] and in the future ITER [2–4]) and the Inertial Confinement Fusion, ICF, approach (on big laser installations like NIF, LMJ, SG-III) [5, 6]. These have resulted in outstanding performance and results, in particular achieved during last two years. In 1999 at tokamak JET the total fusion energy compared to the heating energy supplied of $Q \approx 0.67$ over a very short time has been achieved producing 21.7 MJ [7]. The latest (December 2021) JET experiment with deuterium-tritium fuel sustained $Q \sim 0.33$ for 5 seconds producing 59 MJ [8, 9]. In 2021 at the National Ignition Facility were achieved conditions close to ignition. 1.3 MJ of fusion energy was delivered by irradiating a DT capsule with 1.8 MJ of laser energy [10–12]. In 2022, NIF reported production of 3.15 MJ of fusion energy against 2.05 MJ of energy supplied by the laser energy or a gain of 1.53 [10, 13, 14].

Unlike the deuterium tritium reaction, the proton-boron (pB) fusion reaction [15–20]:



has the very attractive characteristics of not producing neutrons. The absence of neutrons from the primary reaction implies that there will be little activation of materials in a potential reactor and hence a very low amount of radioactive waste. Therefore, the pB fusion reaction is very clean and ecologically acceptable.

In addition, the fact that the reaction produces only charged particles (He^4 nuclei, i.e. α -particles) could imply the additional potential advantage of allowing direct conversion of the produced energy from kinetic to electrical energy, without implying a thermodynamic cycle, hence the traditional production of electricity by turbines (as in current power plants, including fission reactors, but also future fusion reactors based on DT fusion). In principle, this may increase the efficiency of electricity generation [21].

Unfortunately, the pB reaction requires high temperatures to be thermodynamically triggered in a laboratory plasma [22, 23], which explains why in the last decades most of the fusion research has focused on the DT reaction, thus leaving pB fusion as a remote, although interesting, “second step” in production of energy through nuclear fusion. Nevertheless, experimental research on pB fusion as an energy source is still actively investigated by a few laboratories and even by private companies,¹ but the performances in terms of number of triggered fusion reactions are, at the moment, at least two generations behind the studies performed within the standard approaches of MCF and ICF [24].

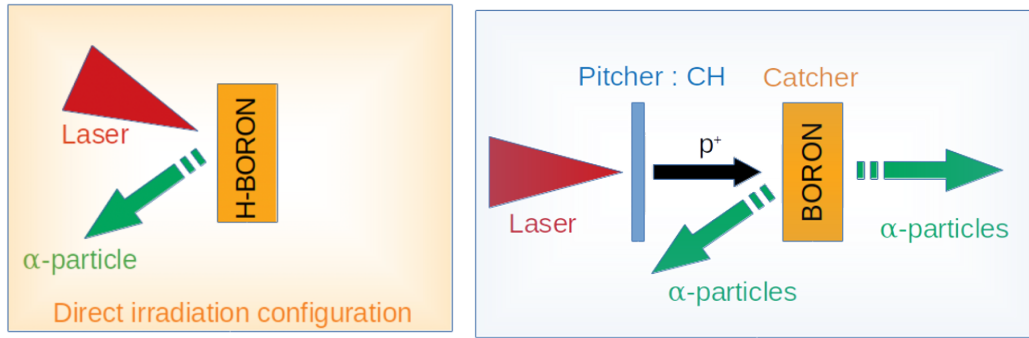


Figure 1. Two mainstream proton boron fusion schemes: (left) in-target irradiation, (right) pitcher-catcher configuration.

However, several experiments recently conducted with laser-produced plasmas have revived the topic of pB fusion. Exciting results have been obtained with two different, mainstream hydrogen boron fusion schemes: in-target irradiation and configuration pitcher-catcher (figure 1).

The in-target irradiation scheme is based on direct interaction of the laser beam with the boron target (containing hydrogen impurities) [25–28]. Here, both boron and hydrogen nuclei are accelerated by various mechanisms (including laser hole boring) to finally react releasing the three α -particles.

In the pitcher-catcher scheme laser irradiates pitcher (usually Al or plastic thin target) producing proton beam, which is sent directly on catcher. Such accelerated, highly energetic protons are directed towards a secondary boron target where the pB reactions take place [29–32].

Most of the experiments within this approach were based on the use of high-energy, high-power laser beams to produce a bright source of protons through the mechanisms of Target Normal Sheath Acceleration (TNSA) [33–37].

Within both approaches, laser-driven experiments have shown unexpected high yield in α -particles production and have then reopened the interest towards such hydrogen-boron fusion.

2 State of art of the proton boron fusion experiment

Historically, the first of this new series of experiments was conducted in 2005 by Belyaev et al. [18] demonstrating the possibility to trigger the pB fusion reaction by using an intense ps laser beam ($2 \times 10^{18} \text{ W/cm}^2$) interacting with a boron-rich polymeric target. In this experiment, protons

¹In particular ENN in China, HB11 Energy in Australia, Marvel Fusion in Germany, TAE Technologies (formerly Tri Alpha Energy) in U.S.A.

generated in the laser-target interaction and hitting B atoms in the bulk of the target, induced the nuclear reactions. An α particle yield of about 10^3 α /sr/shot was estimated in this experiment, later corrected by Kimura et al. [19] to a final yield of 10^5 α /sr/shot. In 2008 Bonasera et al. [38] detected $\approx 10^4$ α /sr/shot in experiments performed at the ABC laser facility in Italy with 1054 nm wavelength, 2.5 ns pulse duration, energy of several-tens of joules and intensity higher than 10^{15} W/cm² on boron-doped plastic targets.

In 2013, Labaune et al. [29] published results of an experiment performed using the LULI laser system in France in which two laser beams were used: the first having ns pulse duration, to ionize a solid boron target, the second having shorter duration (ps regime) but very high laser intensity (6×10^{18} W/cm²) to accelerate the proton beam from a second target. By colliding such laser-accelerated proton beam with the laser-generated boron plasma, they demonstrated a maximum α particle yield of about 10^7 α /sr/shot. Almost simultaneously, Picciotto et al. [25] with a simpler setup to that of ref. [29] and using a sub-ns laser with modest intensity (10^{16} W/cm²) at the Prague Asterix Laser System (PALS) in the Czech Republic, obtained a higher amount of α particles (around 10^9 α /sr/shot). In particular, they used hydrogenated silicon targets, doped by boron through ion implantation technique and enriched with hydrogen by an annealing process.

Finally, in recent papers, using the in-target scheme, Giuffrida et al. [27] reports evidence of proton cutoff at 10 MeV and production of alpha particles well above 10^{10} α /sr/shot on the target front side (i.e. towards the laser) of BN targets irradiated at laser intensity of about 3×10^{16} W/cm² at PALS.

Margarone et al. [31] and later Bonvalet et al. [32] reported the highest (10^9 α /sr/shot) alpha particle flux measured by now in pitcher-catcher configuration at the high-intensity, high-energy Petawatt laser LFEX of Osaka University in Japan. Margarone et al. [28] reported even higher yields of α particles, well above 10^{10} α /sr/shot, using boron nitride (BN) targets and the in-target configuration at LFEX.

A historical perspective of the progression of results is given in figure 2.

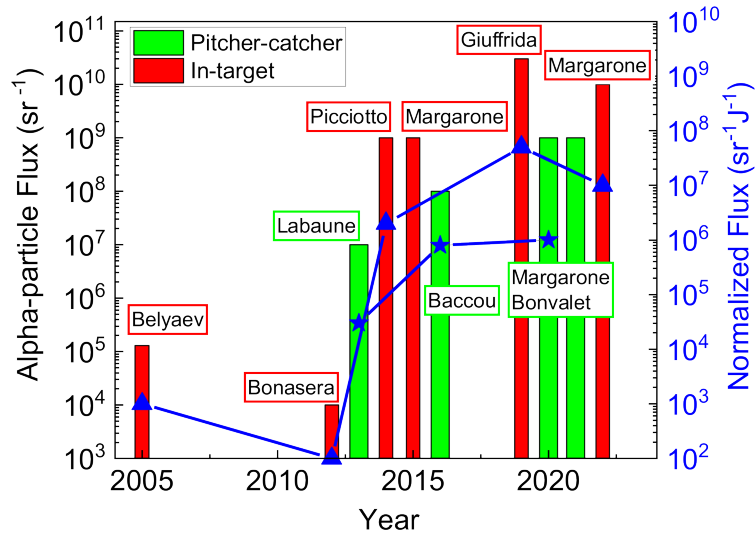


Figure 2. Time evolution of the results from laser-driven proton boron fusion experiments. Reproduced from [28]. CC BY 4.0.

In the experiment at PALS [27] the spectral and angular distribution of emitted α -particles were also characterized, thus, providing the first evidence of laser-induced acceleration in the α spectrum. According to experimental measurements and theoretical calculations two maxima corresponding to alpha particle emission at different energies have been observed. The experimental signal detected with TOF diagnostics have evidenced a dominant emission of protons at 0 degrees. The strongest emission of alpha particles has been observed at bigger angles (e.g. 66 degrees).

The acceleration of alpha particles has been confirmed in the experiment conducted at LFEX [31]. The particularity of this experiment was that lasers like LFEX allow to produce very energetic protons. Thanks to the increased proton penetration range, thicker boron targets could be used in a pitcher-catcher geometry, and α -particles were observed also on the rear side of the boron target. In addition, due to the direct transfer of energy from protons to the reaction products, α -particles with higher energies were produced. This opens the possibility of triggering reactions requiring high-energy α -particles, useful, for instance, for producing radioisotopes of medical interest.

Nowadays, several other experiments are going on worldwide, performed both by research groups taking part in the PROBONO COST Action [39] and by others. Findings look very promising, possibly in view of future feasibility of laser-driven pB fusion as an energy source, or on a shorter timescale in view of the development of new types of compact high-brightness α -particle sources.

Several experimental results also suggest that the obtained high α -particle yields, could also be due to the deviations from thermal equilibrium in laser-plasma experiments, as compared for instance to long-living in-equilibrium magnetically confined plasmas, or even to classical ICF experiments. Hydrodynamic simulations by Hora et al. [40, 41] show that the ignition of a fusion flame could be possible in solid-density $H^{11}B$ fuel irradiated with a high-contrast laser pulse in the ps-PW regime under the effect of magnetic confinement at field strengths of the order of 10 kTesla. Although these must be considered as very preliminary and visionary results, they confirm the clear scientific interest in continuing the investigation of laser-driven pB fusion.

Recently new experiments have been performed in the field of proton boron fusion. Experiments in direct irradiation configuration with boron nitride have been performed at the LFEX / Gekko laser facility with magnetic fields generated using capacitor-coil targets driven by lasers [42]. Experiments to measure the equation of state of boron nitride at high pressures are also underway, in particular at the PALS [43, 44], and ELI-beamlines laser facilities in Prague [45]. These will allow collecting data useful for designing implosion experiments using boron targets for the future production of energy.

Apart from using kJ-class lasers developed for studies related to inertial confinement fusion, rapid progress has also been made in the use of smaller laser systems which might be more adapted for future practical applications related to α -particle sources, including medical applications. The high-repetition-rate short-pulse PW laser at CLPU in Salamanca, Spain, has been used to assess the capability of producing high average fluxes of α -particles and to test the possibility of characterizing α -particle generation using nuclear-based diagnostics [46]. In another experiment [47] a tabletop laser system with low energy (~ 10 mJ) and peak power ~ 10 GW) has provided 6×10^4 α particles/s at 10 Hz and 10^6 α -particles/s at 1 kHz.

3 Current challenges

The experiments on pB fusion with lasers are still in their infancy: despite the first very positive and encouraging results there are still substantial theoretical and experimental investigations to be carried out both in terms of fundamental science and applied research. In order to progress beyond the state of the art, it is possible to use the wide “laser science park” created in recent years, by facilitating access to research infrastructures. In particular in Europe, the start of operation of new facilities such as ELI [48–50] coupling high-repetition rate and high-pulse-energy, enormously extends the parameter space available for studies, allowing to perform novel experiments. In particular, the L4 laser systems at the ELI-beamlines center near Prague offers the capability of shooting kJ laser energy at higher repetition rates (~ 1 shot per minute) which could never be previously achieved. A 2PW laser system operating at 10 Hz is already available at ELI-NP in Romania. In general, laser *R&D* is rapidly progressing towards providing multi-PW powers at high repetition rates, a development which clearly fits very well with the needs of research on proton boron fusion and applications.

In addition, it is necessary to work with laboratories, research groups, and companies from materials science to study and optimize new type of boron targets allowing increased α -particle yield. In particular, the recent developments on micro and nano technologies are extremely promising and could be very useful for proton boron fusion research.

The scientific challenges can be answered only through the establishment of efficient and effective networking. This is indeed the goal of the COST Action 21128 “PROton BORon Nuclear fusion: from energy production to medical applicatiOns” PROBONO [39] by supporting Short-Term Scientific Missions (STSMs) and by organizing dedicated workshops and schools addressed in particular to Ph.D. students and young post-docs.

The key is to share knowledge, resources, promoting global research cooperation and access to laboratories where experiments can be carried out. To be able to cover a variety of different plasma conditions, several different laser drivers must be used. Beyond realization of experiments, study of new type of targets, the PROBONO Action addresses three specific challenges related to development of simulations [51], diagnostics and targetry.

3.1 Challenge on development of novel diagnostics

The detection of the ionic products of low-rate fusion reactions, and in particular of the pB fusion reaction, is one of the recognized problems in experiments [52]. Detection of α -particles is intrinsically challenging in high power laser-plasma experiments, because of the simultaneous presence of several other accelerated plasma ions. Moreover, the spectrum of α -particles reaching a detector depends not only on the implemented experimental scheme, but also on the angle of detection. In fact, α -particles will interact with the solid target and/or with the surrounding plasma, and this will produce modification of the detected spectrum (as, i.e. shown in [19]). In the pitcher-catcher configurations [32], momentum conservation can lead to heavy modifications of the emitted α -spectrum showing higher energy due to direct transfer of energy from protons to the reaction products.

Until now, the fundamental method of detection for α -particles in this type of experiments has been the use of Solid-State Nuclear Track Detectors (SSNTD), where exposition to ionizing radiation generates local damaging [52, 53]. Typical examples are the CR39, and the PM-355 plastic polymers, where damages are caused by the breaking of the long polymer chains due to incoming radiation.

Along these damaged regions, the material is more susceptible to chemical attack and, tracks become visible with sizes increasing with the duration of the etching procedure. These can be properly characterized by confocal microscope imaging providing information on particle energy and on type. The problem is that, besides α -particles, the detector is also reached by all other plasma ions, which are of much higher number. Discrimination between α particles and other particle species can be (partially) achieved by using filters. In some cases, the shape of the etched tracks can also be used as a discrimination factor [54]. Automatic recognition systems have already been developed, for instance the system Neutrak© by Landauer [55, 56], which uses a deterministic image analysis algorithm to separate neutron tracks (the object of interest in that case) from impurities / background / α -particle tracks and other heavy charged particle tracks. The first attempt of going beyond this stage by applying advanced machine learning capabilities so to automatically classify particle tracks in CR39 and differentiate between different particles and different energies has been made by Amit, Nissim et al. [57]. Detection of α -particles is difficult in Thomson parabola spectrometers [52, 58, 59] because their trace is superimposed to those of ions with the same charge/mass ratio (0.5), such as fully stripped C, O, B, etc. For example, α -particles will have the same position as $^{12}\text{C}^{6+}$ ions with three times bigger energy. In such cases, a discrimination might be performed by using SSNTD detectors, since traces may have fairly different diameters [60] and again putting filters in front of the SSNTD.

Time-Of-Flight schemes may be applied to the detection of fusion products in pB reaction. However, TOF give no discrimination on particles but only on their velocities. Thus α -particles and other ions (mainly protons) can simultaneously arrive to the detector whenever the spectrum of protons or of α -particles are non-monoenergetic. One possible way to use the TOF technique effectively is to couple it to a suitable Thomson spectrometer. However, these methods present the drawback of the small acceptance solid angle, which implies reduced sensitivity.

Due to the limitation of state-of-the-art diagnostic approaches, it is necessary to develop innovative new diagnostic methods [19, 58–61], or modify and upgrade the existing solutions. For instance, it will be possible to get information on pB reactions from studying other reactions occurring in the same experiment, and secondary generating products that can be detected more easily. As an example, ^{11}B is present in about 80% of natural B, the remaining being ^{10}B . In experiments with natural B, the reaction $^{10}\text{B}(p, \alpha)^7\text{Be}$ will take place simultaneously to the expected $^{11}\text{B}(p, \alpha)2\alpha$, even though with a smaller cross section. The released radioactive ^7Be decays emitting 477 keV γ -rays. It can be collected from the residues of the target after the interaction and can be detected more easily than α -particles, for example by high sensitivity liquid nitrogen cooled germanium γ -detectors. This allows to retrieve the number of pB reactions by knowing the ratio $^{10}\text{B}/^{11}\text{B}$. This method is insensitive to target thickness, i.e. to the fact of whether the α -particles can escape from the target or not. It can then be used for estimation of the total number of occurred fusion reactions. First applications of this method are found in [62, 63]. Another useful reaction is $^{10}\text{B}(p, \gamma)^{11}\text{C}$, releasing radioactive ^{11}C , which normally decay with positron emission, $^{11}\text{C} \rightarrow ^{11}\text{B} + e^+ + \nu_e + 0.96 \text{ MeV}$. Again, γ -emission at 512 keV can be an important observable. Alternatively, the interaction of protons and ^{11}B can generate ^{11}C and neutrons through the nuclear reaction: $p + ^{11}\text{B} \rightarrow n + ^{11}\text{C} - 2.9 \text{ MeV}$. Therefore, a neutron diagnostic could also allow to estimate the number of protons with energy higher than $\approx 3 \text{ MeV}$ reaching the target in some configurations. At the same time, we can observe the production of excited F^* and N^* with lower rate. This could also open the possibility for additional diagnostic channels.

In conclusion, in order to accurately measure the characteristic of emitted α -particles, a simultaneous use of different techniques is needed. Thus, a substantial development work must be done and the progress will clearly benefit from a coordinated action. In order to have suitable insight on the actual mechanisms producing laser-generated α , it is of primary importance to have also a very accurate characterization of the interaction in both the pitcher-catcher and direct laser-interaction schemes.

3.2 Challenge on targetry

While experiments using high energy lasers are basically “single shots”, the development of α -particle sources for applications requires the use of high-repetition-rate (HRR) lasers. In this case, efficient technologies for fabrication and metrology of targets in large quantity need to be developed, at high precision and accessible cost. We need easy-to-align and cheap targets, which can be positioned in the chamber without breaking the vacuum. This relies on using engineered holders containing many targets, positioning them with high precision, and at the same time avoiding/shielding target debris after each shot. This work will benefit from already-collected experience of operating on lower-energy HRR lasers and XFELs.

At the same time, we need to develop and characterize new types of targets to optimize α -particle production. This will need the synergic work of experimentalists and material scientists. Targets may be structured on the front surface in order to maximize laser absorption, they may contain specific layers in order to tailor the generation of α -particles, may be enriched in hydrogen. Advanced approaches will include hydrogenated BN nanoparticles, hydrogenated BN nano-sheets, hydrogenated BN nano-fibers [64] on a plastic substrate, possibly using a transfer process to the substrate by Laser Induced Forward Transfer technique (LIFT) as already demonstrated for carbon nanowalls [65, 66]. As written before, the recent developments on micro and nano technologies are extremely promising and could be very useful for proton boron fusion research. Recently the use of nano-structured materials has been proposed as the basis of a new reactor concept, being potentially capable of driving ion beams with very high efficiency [67].

3.3 Challenge on development on adapted numerical simulation tools

The use of laser beams to trigger pB fusion reactions requires a chain of numerical codes for interpreting and describing all involved physical processes. The main process is the production of energetic protons by lasers inducing the generation of α -particles in a secondary boron target, or directly in a hydrogenated boron-rich target. In experiments with high intensity laser systems, the main laser pulse is preceded by a low intensity pedestal, produced by amplified spontaneous emission. This prepulse may induce target expansion, pre-plasma formation, and a shock wave propagating through it, changing the density, ionization and temperature of the material. These changes modify the interaction of the main laser pulse with the target and consequently the production of protons.

Simulations of prepulse effects can be performed with 2D hydrodynamics codes (e.g. [68]) provided they contain the relevant physics packages related to interaction of low and intermediate laser intensities with matter, ion and electron heat conduction, thermal coupling, detailed radiation transport, and realistic equation of state allowing the calculation of the state of matter and its time evolution.

Above a given laser intensity, typically $\approx 10^{16}$ W/cm², other processes occur, and different types of codes must be used such as Particle in Cell (PIC) codes (e.g. SMILEI [69] or WARPX [70])

which compute laser matter interaction at high intensities. Both SMILEI and WARPX are open codes used to model many plasma physics scenarios. SMILEI is a collaborative project providing the scientific community with an open-source, user-friendly, high-performance and multi-purpose code for plasma simulation. SMILEI and WARPX simulations may be initialized with the using the temperature and density profile outputs from hydrocodes following the effects of the laser prepulse. Then they may compute proton production vs. energy, angle and position.

Finally, Monte-Carlo (MC) codes, like FLUKA [71] or GEANT4 [72], can model the interaction and the transport of ions, electrons, photons and neutrons. FLUKA is continuously benchmarked against models and experimental data of nuclear cross sections and specifically the pB fusion reaction [73].

This chain must be optimized in order to allow the simulation codes to communicate with each other in a fast and reliable way. The phase space of protons calculated by a PIC code can initiate a MC simulation to obtain the α -particle yield at the end of the code chain. Each code of the chain must be improved in physics or in numerics, for instance by extending the hydrocode to 3D, by implementing more accurate physics package for collisional processes and faster numerical schemes for PIC code. Furthermore, new trends on PIC simulations using GPU's on HPC's (e.g. with PICongPU) have been accomplished minimizing the runtime of the simulations [74] and will be used for large scale models (2D and 3D).

For MC code, it will be necessary to implement cross sections specifically developed for the case of warm dense matter, to consider the heating of the target. Although some work on adapting MC codes to plasma conditions has already been done, clearly much works remain [75]. Complementarily, the development of Fokker-Planck hybrid modeling for plasma targets will allow more precise calculations of the stopping power by including plasma effects.

4 Summary

Recent results achieved in the field of proton boron fusion has attracted a lot of attention of the academic community and private investors.¹ It is certainly worth citing the fact that, in parallel to laser-driven proton boron fusion, several other concepts leading to proton boron fusion are developing. These include the use of reversed field pinch, stellarators, plasma focus devices, vacuum discharges, etc. [76–79]. New findings and promising results stimulate the growth of the community [80–82]. It requires technological progress allowing the production of new materials, detectors and data analysis techniques. The PROBONO action aims at promoting and coordinating the work on the subject, thus accelerating the exchange of knowledge, increasing motivation and allowing for faster progress in proton boron fusion, both for the future goal of energy production and for shorter-time applications.

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